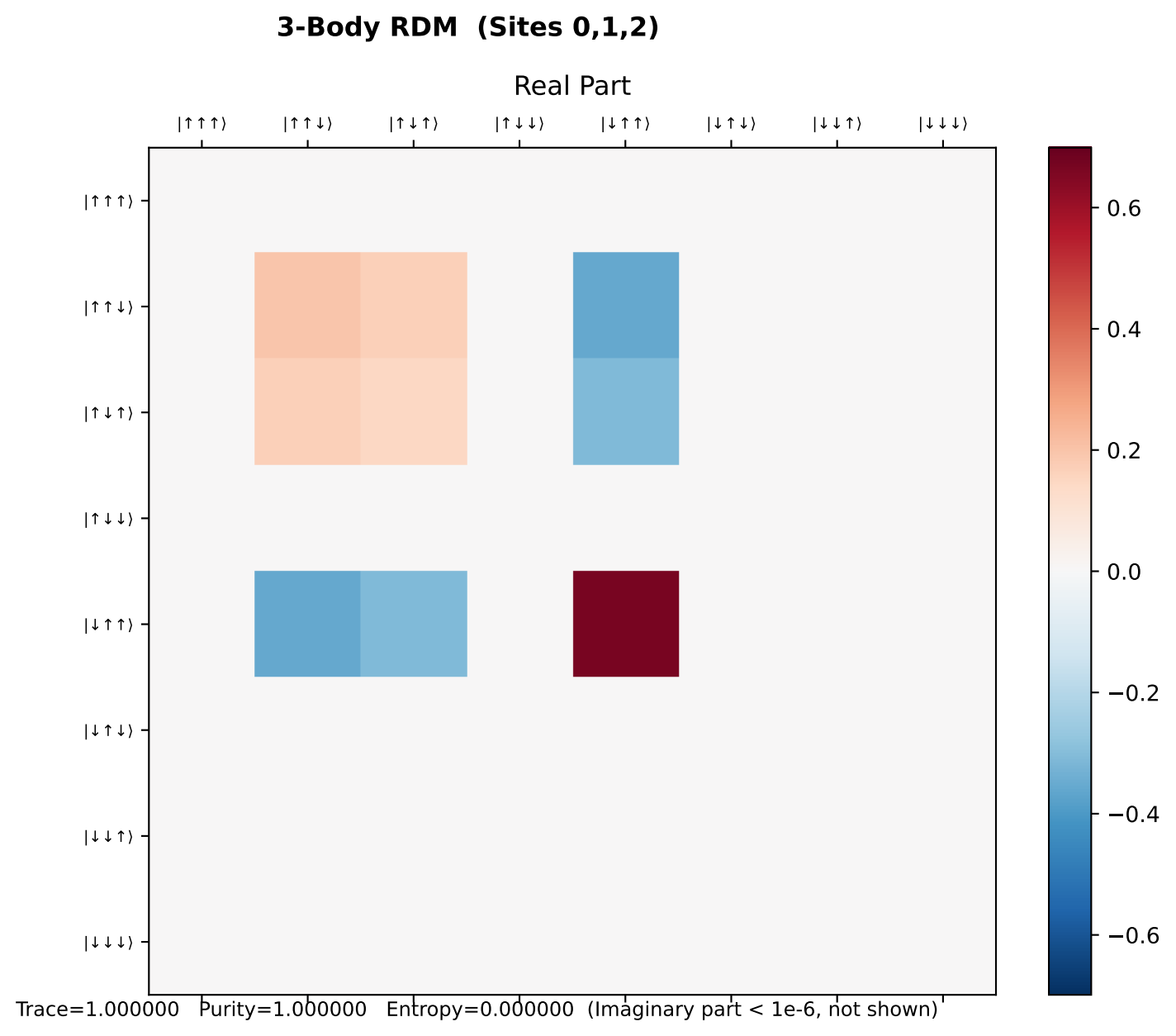
Genuine Multipartite Entanglement Analysis

```
Model                              Ground-state properties           GME summary
-----                              ----------------------           -----------
Hubbard ring (1-D, PBC)            E0                : -1.27491722  Extracted n-spin RDM : n = 3
                                   E1                : -1.27491722
System parameters                                                   Purity:
-----------------                  <S2>              : 0.750000       gamma = Tr(rho^2)
Sites (L)          : 3             S                 : 0.5000
Electrons (Ne)     : 3             <Sz>              : 0.250000     Separable cut criterion:
Hopping (t)        : 1.0           Multiplicity 2S+1 : 2.0            gamma_A = 1  <->  rho = rho_A x rho_B
Interaction (U)    : 4.0           Spin state        : Doublet (S=1/2)
Mode               : uniform                                       C_GME (Troiani et al.):
n-body order       : 3             Double occupancy                   C_GME = sqrt(2 * min_A (1 - Tr(rho_A^2)))
                                   ----------------
                                   <D> total         : 0.230835     Convex roof (mixed):
                                   <D> / site        : 0.076945       C_GME(ρ) = min Σ pᵢ C_GME(|ψᵢ))


                                                                    Date: 2026-04-10  21:07
```


3-Body RDM (Sites 0,1,2)
Real Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 3-Body RDM (Sites 0,1,2)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 1.0000000000
  Von Neumann entropy  : 0.0000000000
  Eigenvalue min       : -1.673436e-16
  Eigenvalue max       : 1.000000e+00
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 1.00000000

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------
  (0,)         | (1, 2)       | 0.55479385
  (0, 1)       | (2,)         | 0.69076186
  (0, 2)       | (1,)         | 0.75444429

STATUS: GME EXISTS
------------------------------------

C_GME = 0.70079342
GME confirmed across ALL cuts.
```

```
Convex Roof Analysis
====================================

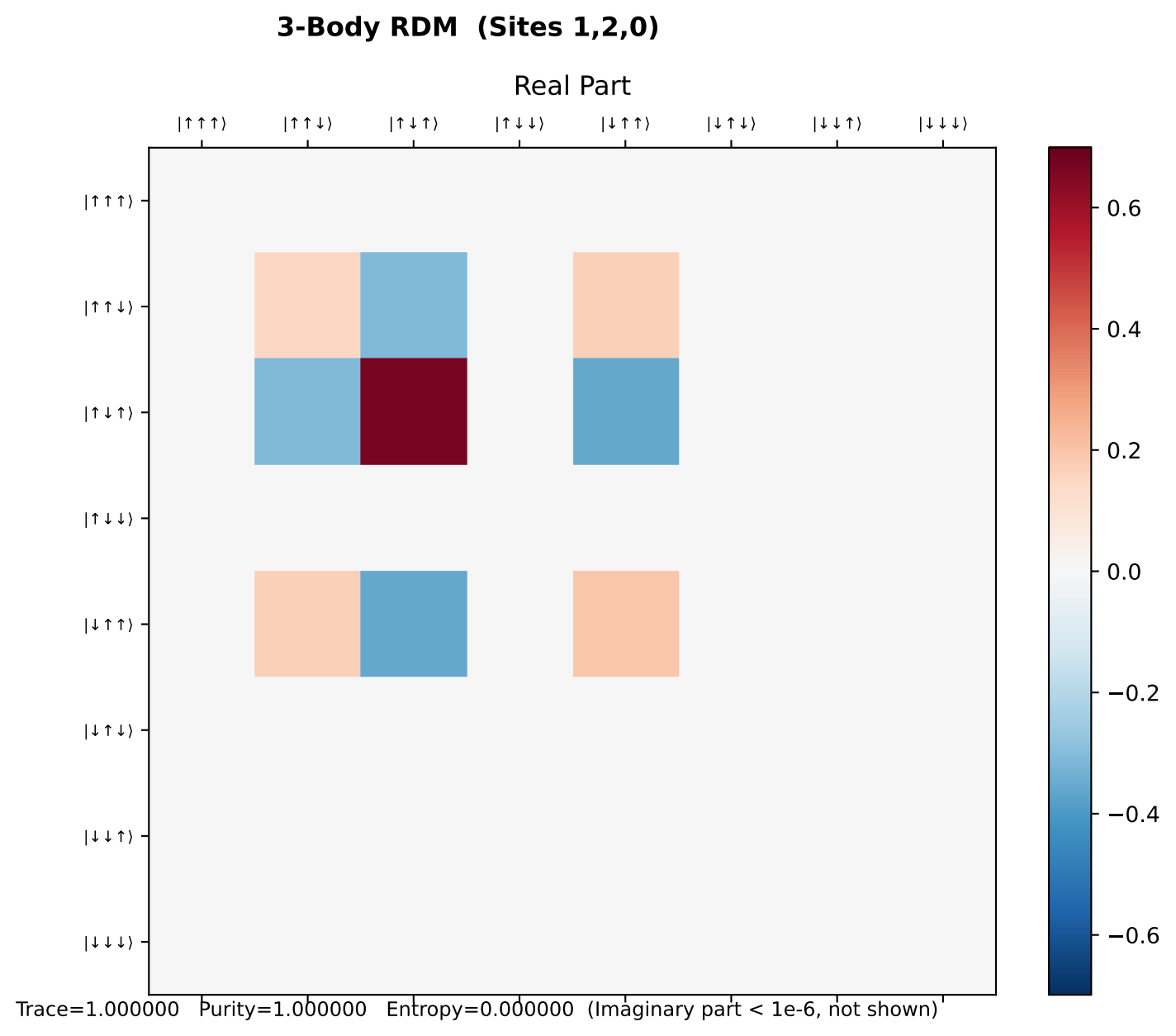
  Lower Bounds:
    Paper (witness)     = 0.00000000
    Notes (6-9)    = 0.00000000
    Best C_LB           = 0.00000000

  C_GME (unitary optimization - Cayley Transform):
    C_num (unitary opt) = 0.70079342   BEST
    (20 restarts)

  C_GME (unitary optimization - Expoenetial Map):
    C_num  = 0.70079342   BEST
    (L-BFGS-B, 20 restarts)

  Upper Bound:
    C_UB (eigendecomp)  = 0.70079342

  Status: GME
  Bounds consistent: YES
  Convergence: TIGHT
```


3-Body RDM (Sites 1,2,0)
Real Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 3-Body RDM (Sites 1,2,0)

```
RDM Metrics
==================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 1.0000000000
  Von Neumann entropy  : 0.0000000000
  Eigenvalue min       : -3.112091e-17
  Eigenvalue max       : 1.000000e+00
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 1.00000000

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------
  (1,)         | (2, 0)       | 0.69076186
  (1, 2)       | (0,)         | 0.75444429
  (1, 0)       | (2,)         | 0.55479385

STATUS: GME EXISTS
------------------------------------

C_GME = 0.70079342
GME confirmed across ALL cuts.
```

```
Convex Roof Analysis
==================================

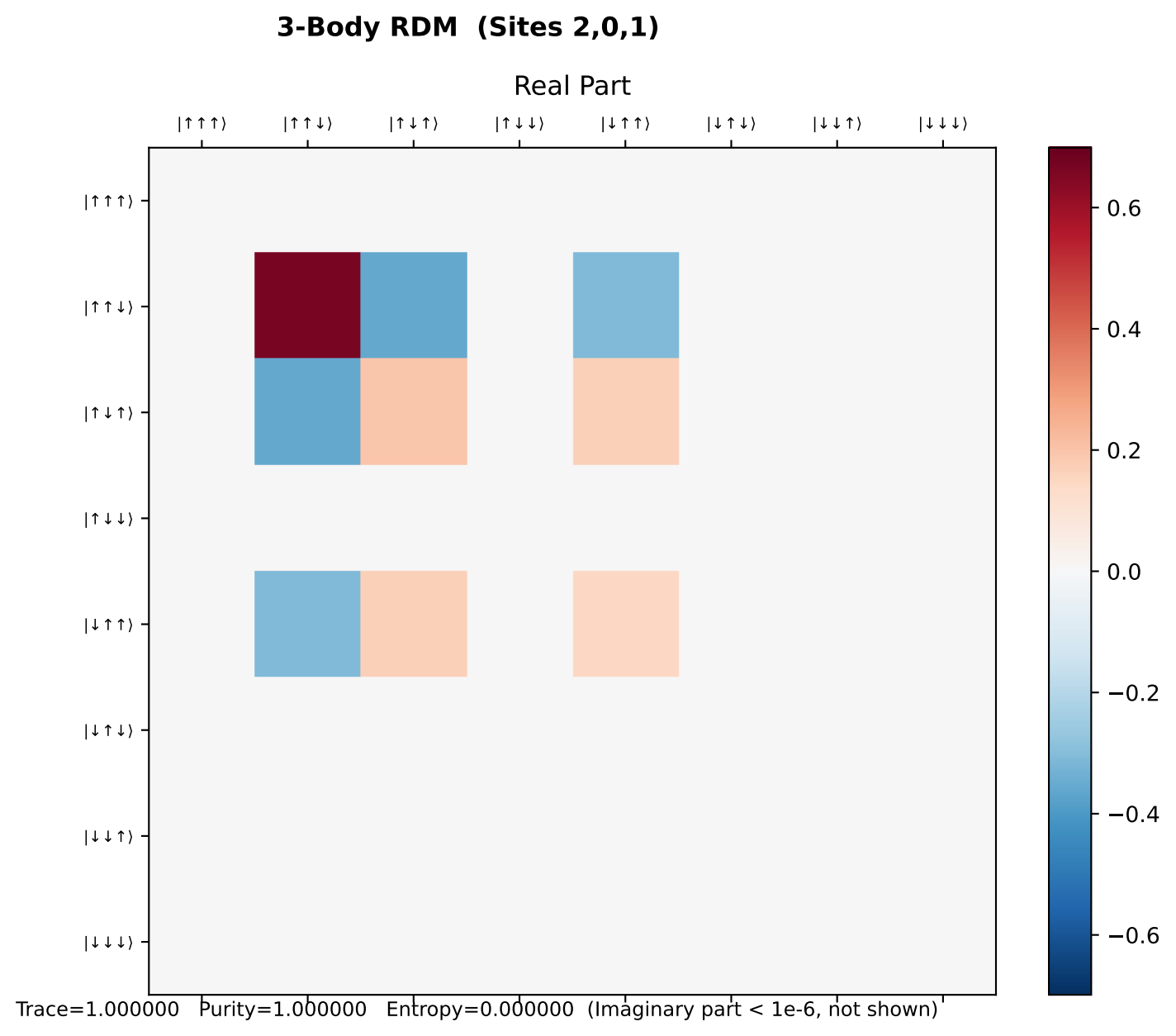
  Lower Bounds:
    Paper (witness)     = 0.00000000
    Notes (6-9)    = 0.00000000
    Best C_LB           = 0.00000000

  C_GME (unitary optimization - Cayley Transform):
    C_num (unitary opt) = 0.70079342   BEST
    (20 restarts)

  C_GME (unitary optimization - Expoenetial Map):
    C_num  = 0.70079342   BEST
    (L-BFGS-B, 20 restarts)

  Upper Bound:
    C_UB (eigendecomp)  = 0.70079342

  Status: GME
  Bounds consistent: YES
  Convergence: TIGHT
```


3-Body RDM (Sites 2,0,1)
Real Part
|↑↑↑⟩
|↑↑↓⟩
|↑↓↑⟩
|↑↓↓⟩
|↓↑↑⟩
|↓↑↓⟩
|↓↓↑⟩
|↓↓↓⟩
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 3-Body RDM (Sites 2,0,1)

```
RDM Metrics
====================================

  Trace(rho)           : 1.0000000000
  Purity Tr(rho^2)     : 1.0000000000
  Von Neumann entropy  : 0.0000000000
  Eigenvalue min       : -1.154627e-16
  Eigenvalue max       : 1.000000e+00
  Imaginary part       : < 1e-6 (not shown)
```

```
GME Analysis
====================================

  gamma = Tr(rho^2) = 1.00000000

  A            | B            | Tr(rho_A^2)
  ------------ | ------------ | -----------
  (2,)         | (0, 1)       | 0.75444429
  (2, 0)       | (1,)         | 0.55479385
  (2, 1)       | (0,)         | 0.69076186

STATUS: GME EXISTS
------------------------------------

C_GME = 0.70079342
GME confirmed across ALL cuts.
```

```
Convex Roof Analysis
====================================

  Lower Bounds:
    Paper (witness)     = 0.00000000
    Notes (6-9)    = 0.00000000
    Best C_LB           = 0.00000000

  C_GME (unitary optimization - Cayley Transform):
    C_num (unitary opt) = 0.70079342   BEST
    (20 restarts)

  C_GME (unitary optimization - Expoenetial Map):
    C_num  = 0.70079342   BEST
    (L-BFGS-B, 20 restarts)

  Upper Bound:
    C_UB (eigendecomp)  = 0.70079342

  Status: GME
  Bounds consistent: YES
  Convergence: TIGHT
```